

Program : Diploma in Renewable Energy	
Course Code : 4299	Course Title:- Energy management and Auditing
Semester : 4	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L: 3 T: 0 P: 0)	Periods per semester: 45

Course Objectives:

- To enable the students to understand the concepts of energy management and energy auditing
- To understand the energy saving opportunities in Thermal utilities
- To understand the energy saving opportunities in Electrical utilities
- To Identify the different methods used for the economic analysis

Course Prerequisites: NIL

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of energy management and energy auditing	10	Understanding
CO2	Outline the energy saving opportunities in thermal utilities	12	Understanding
CO3	Summarize energy saving opportunities in electrical utilities	11	Understanding
CO4	Identify the financial analysis techniques and energy economics	10	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				2		
CO2	2				2		
CO3	2				2		
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of energy management and energy auditing		
M1.01	Explain the basics of energy and its various forms	2	Understanding
M1.02	Outline the general principles and objectives of energy management	3	Understanding
M1.03	Interpret the various steps in energy auditing	3	Understanding
M1.04	Summarize various energy auditing Instruments	2	Understanding
Contents: Basics of energy & its various forms -Classification of energy sources, Past and Present scenario of world, Environmental aspects associated with energy utilization,Energy conservation and its importance, Energy Management -Definition and objective -General principles, Energy Management skills , Energy Management Strategies,-Responsibilities and duties of Energy managers and auditors Energy Audit -definition(as per EC Act)-need of energy audit,types of energy audit - preliminary -targeted - detailed audit.Various stages for preparing energy audit(formation of energy audit team,energy audit forms and data sheets)-various phases of energy audit-ten steps methodology(as per BIS standard), Benchmarking,			

Energy Audit Instruments- Fuel efficiency Monitor, Combustion Gas Analyzers, Manometer with pitot tube, Ultrasonic flow meters, Contact thermometers, Non contact Infrared Thermometer, Stroboscopic non contact Tachometers, Lux meters, PH meters, Power analyzers, Radiation pyrometer. (Basic concept and working Principle)

CO2	Outline energy saving opportunities in thermal utilities		
M2.01	Explain different fuels and combustion process involved in various thermal utilities	2	Understanding
M2.02	Outline energy saving opportunities in Boilers and furnaces	4	Understanding
M2.03	Explain energy saving opportunities in steam system, Insulation and Refractories	3	Understanding
M2.04	Summarize the concepts of cogeneration and waste heat recovery	3	Understanding
	Series Test – I	1	

Contents :

Fuels and Combustion-Types of fuels,, Combustion of fuels-principle-3T's of combustion- combustion controls(various types),

Energy management opportunities in Boilers - Boiler types and classification, Energy conservation opportunities in Boilers,

Energy management opportunities in Furnaces-Types and classification of furnaces, Energy conservation opportunities in Furnaces.

Steam system, Insulation and Refractories -Properties of steam, Energy saving opportunities in steam system, Insulation types and applications, Refractories-examples-general requirements of a refractory material-properties ,classification and selection of refractories

Cogeneration- definition-principle-need-advantages- Cogeneration systems –steam turbine system -gas turbinesystem -reciprocating engine system (schematic diagram representation and brief idea only), Classification based on sequence of energy use

Waste Heat Recovery-Introduction, classification and applications, Benefits of waste heat recovery, Commercial Waste Heat Recovery Devices(short note about - Recuperators,Regenerators,Heat wheel,Heat pipe,Economise,Heat exchanger).

CO3	Summarize energy saving opportunities in electrical utilities		
M3.01	Outline Energy saving opportunities in Lighting systems	3	Understanding
M3.02	Explain energy saving opportunities in electric motors and electrolytic process	3	Understanding
M3.03	Explain the energy saving opportunities in Compressor system and HVAC systems	2	Understanding

M3.04	Summarize the energy saving opportunities in fans, blowers, pumps ,air conditioning system, cooling tower, D.G sets	3	Understanding
Contents : Energy management opportunities in Lighting Systems- Introduction, General energy saving opportunities, Energy efficient lighting controls Energy management opportunities in Electric Motors- Introduction, Energy Efficient Motors, Factors affecting energy efficiency and minimizing motor losses, Energy management opportunities in Electric motor. Energy management opportunities in Electrolytic process and Electric heating. Energy management opportunities in HVAC System & Compressor Systems Introduction, Coefficient of performance(Definition), Energy saving opportunities in Refrigeration plants, Compressed Air system components, Efficient operation of compressed air system. Energy efficiency in Pumps, Fans, Blowers, Air Conditioning System, Cooling Towers, D.G sets.			
CO4	Identify Financial analysis techniques and energy economics		
M4.01	Outline the energy economics	2	Understanding
M4.02	Identify the Financial analysis techniques	4	Applying
M4.03	Explain the Cash flow diagram and Energy saving Company Concepts	2	Understanding
M4.04	Interpret star labeling,needs and its benefits	2	Understanding
	Series Test – II	1	
Contents : Energy Economics -Introduction, Investment need appraisal and criteria, Energy monitoring and targeting,Benefits of monitoring and targeting Financial analysis techniques - Calculation of energy savings , Payback period method, Average rate of return Method, Net present value method, Internal rate of return method(Advantages & Disadvantages of each method), Simple problems using payback period method . Cash Flow and ESCO - Capital Investment considerations, Cash Flow Diagrams, Financing options, Time value of money, ESCO Concept , Energy performance contracting and Role of ESCOS, Types of performance contracting. Star labeling -Need and its benefits,Energy efficiency standards, Energy efficiency labels, types,comparative label,Endorsement label,Minimum Energy Performance Standards(MEPS), Star ratings, List of appliances covered under standards and Labeling program.			

Text / Reference

T/R	Book Title/Author
T1	Guide books for National Certification Examination for Energy Manager / Energy Auditors Book-1, General Aspects
T2	Guide books for National Certification Examination for Energy Manager / Energy Auditors Book-3, Electrical Utilities
T3	Guide books for National Certification Examination for Energy Manager / Energy Auditors Book-2, Thermal Utilities
R1	Wayne C. Turner, Energy management Hand Book - - The Fairmont Press, Inc., 1997.

Online Resources

Sl.No	Website Link
1	http://www.tcci.com.au/Resources/Sustainability/Earn-Your-Stars/The-Process
2	http://www.opexworks.com/KBase/Energy_Management/Energy_Audit_and_Management/Energy_Audit_Instrumentation_and_Control/Energy_Audit_Instruments.htm
3	https://www.energysage.com/energy-efficiency/101/ways-to-save-energy/